

WAYNE COUNTY AP, OH, USA

WMO: 725216

Lat: 40.873N

Lon: 81.887W

Elev: 1137

StdP: 14.10

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 04842

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.1	5.7	-6.3	4.1	1.0	-1.0	5.4	7.1	26.6	29.0	24.6	29.9	5.5	280	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	88.1	72.8	85.7	71.9	83.4	70.6	75.7	83.9	74.3	82.0	72.8	80.1	9.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.0	127.9	80.4	71.9	123.1	78.9	70.4	116.7	77.1	39.9	84.3	38.5	81.9	37.1	79.9	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.4	19.9	17.8	-7.4	90.9	6.9	2.0	-12.3	92.4	-16.3	93.6	-20.2	94.7	-25.2	96.2	
			DB												
			WB	-7.9	78.3	6.6	1.6	-12.7	79.5	-16.6	80.4	-20.3	81.4	-25.1	82.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.0	25.8	28.1	37.5	49.7	60.2	68.5	71.5	69.3	63.2	52.2	41.2	31.4		
	DBStd	18.40	12.25	11.52	11.15	9.22	8.31	6.33	5.05	4.92	6.94	8.74	9.19	10.26		
	HDD50	2863	753	616	415	117	12	0	0	0	2	77	290	581		
	HDD65	6100	1216	1032	853	464	196	38	7	17	114	407	714	1042		
	CDD50	2868	2	5	27	108	329	556	667	600	397	146	27	4		
	CDD65	627	0	0	0	5	48	144	209	151	59	11	0	0		
	CDH74	5394	0	0	7	75	479	1180	1754	1216	583	99	1	0		
	CDH80	1420	0	0	0	8	102	337	517	285	154	17	0	0		
	WSAvg	7.7	10.0	9.6	9.3	9.4	7.7	6.9	5.5	4.8	5.1	6.7	8.3	9.2		
	PrecAvg	37.0	2.2	1.9	2.7	3.3	4.0	3.7	3.9	3.9	3.3	2.5	3.0	2.6		
Precipitation	PrecMax	53.3	6.5	4.6	6.5	6.3	8.3	8.7	12.8	9.4	8.5	5.9	9.4	6.9		
	PrecMin	22.5	0.6	0.2	0.8	0.8	1.1	0.7	0.7	0.5	0.6	0.1	0.5	0.6		
	PrecStd	6.6	1.3	0.9	1.1	1.4	1.7	1.9	2.5	1.8	1.9	1.3	1.5	1.2		
	PrecStd	6.6	1.3	0.9	1.1	1.4	1.7	1.9	2.5	1.8	1.9	1.3	1.5	1.2		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.5	64.7	73.0	81.0	86.3	90.2	90.8	89.3	88.6	82.3	70.0	60.6		
		MCWB	55.2	56.0	60.0	63.9	69.5	71.7	74.9	74.0	72.1	69.1	58.3	55.1		
	2%	DB	54.3	56.0	65.8	75.6	82.8	87.2	88.0	85.9	84.5	76.3	64.1	55.3		
		MCWB	51.1	51.3	56.2	60.2	67.2	71.2	74.1	73.1	70.8	63.9	56.2	51.0		
	5%	DB	48.2	50.4	60.8	71.6	79.8	83.9	85.4	83.1	80.9	72.0	60.8	51.1		
		MCWB	44.9	46.1	52.9	58.3	66.2	70.0	72.7	71.6	69.1	61.7	53.9	47.6		
	10%	DB	42.8	44.3	55.4	66.2	75.5	81.1	82.5	80.9	76.8	67.3	56.1	46.0		
		MCWB	39.6	39.9	49.6	55.7	64.3	68.8	71.1	70.2	66.3	60.2	50.5	43.1		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.0	57.5	61.7	66.5	72.9	75.4	78.3	77.1	75.5	71.2	62.1	56.3		
		MCDB	59.3	63.8	71.7	76.4	82.7	84.9	87.3	85.3	84.3	80.0	67.6	59.2		
	2%	WB	51.7	52.0	57.8	63.0	70.2	73.5	76.3	75.2	73.0	67.4	57.8	52.2		
		MCDB	54.3	55.0	64.5	72.3	79.0	82.4	84.6	82.6	80.5	72.7	62.5	54.7		
	5%	WB	45.8	46.1	54.3	60.2	68.0	72.1	74.7	73.4	70.8	63.9	54.7	47.8		
		MCDB	48.0	50.1	59.7	68.6	76.5	80.8	82.3	79.9	77.6	69.1	59.4	50.6		
	10%	WB	39.5	39.9	50.0	57.2	66.0	70.5	73.0	72.0	68.7	60.5	51.1	43.2		
		MCDB	42.5	43.9	55.4	64.6	73.5	78.2	80.0	77.9	74.7	66.5	55.9	46.4		
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.7	17.5	21.2	21.2	19.6	19.9	20.6	22.2	20.4	16.8	13.0		
		MCDBR	18.8	22.0	25.1	27.5	25.5	23.3	22.1	23.0	26.1	26.4	23.6	18.3		
		MCWBR	16.1	17.5	16.6	16.0	12.8	10.8	10.3	11.9	13.7	15.1	16.5	15.7		
	5% WB	MCDBR	17.4	21.0	23.0	23.7	21.5	19.7	18.6	18.6	21.4	21.4	20.3	17.8		
		MCWBR	16.3	17.9	16.3	15.6	12.1	10.3	9.4	10.6	12.0	13.8	15.8	16.7		
Clear-Sky Solar Irradiance	taub		0.300	0.328	0.351	0.386	0.422	0.444	0.444	0.431	0.379	0.360	0.320	0.303		
	taud		2.424	2.329	2.351	2.280	2.209	2.186	2.205	2.227	2.350	2.396	2.475	2.451		
	Ebn at Noon		272	279	285	281	272	265	264	264	272	264	263	261		
	Edh at Noon		25	32	36	41	45	46	45	43	35	30	24	23		
All-Sky Solar Radiation	RadAvg		464	737	1057	1439	1712	1876	1871	1689	1361	898	584	390		
	RadStd		42	96	104	133	144	140	121	81	122	78	85	43		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+137
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+191

Nomenclature: See separate page